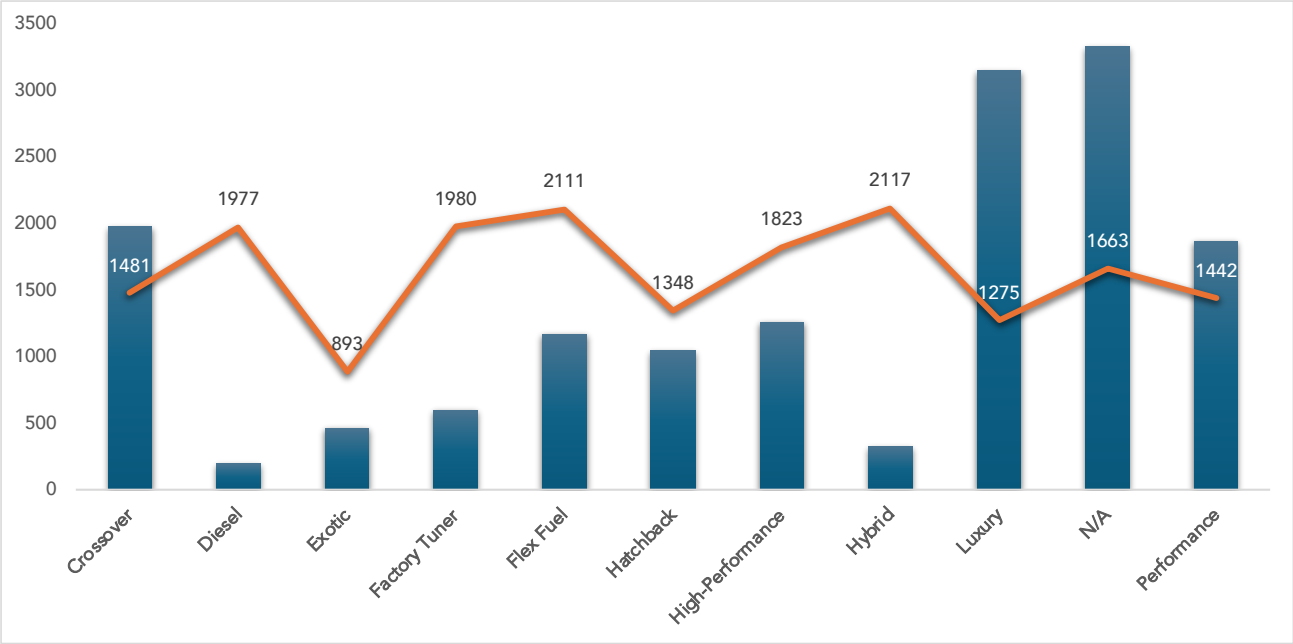
Analyzing the Impact of Car Features on Price and Profitability

Insight Required: How does the popularity of a car model vary across different market categories?

- **Task 1.A:** Create a pivot table that shows the number of car models in each market category and their corresponding popularity scores.
- **Task 1.B:** Create a combo chart that visualizes the relationship between market category and popularity.

ANS.

Market Category	Count of Model	Average of Popularity
Crossover	1983	1481
Diesel	201	1977
Exotic	472	893
Factory Tuner	604	1980
Flex Fuel	1173	2111
Hatchback	1046	1348
High-Performance	1261	1823
Hybrid	330	2117
Luxury	3155	1275
N/A	3338	1663
Performance	1871	1442



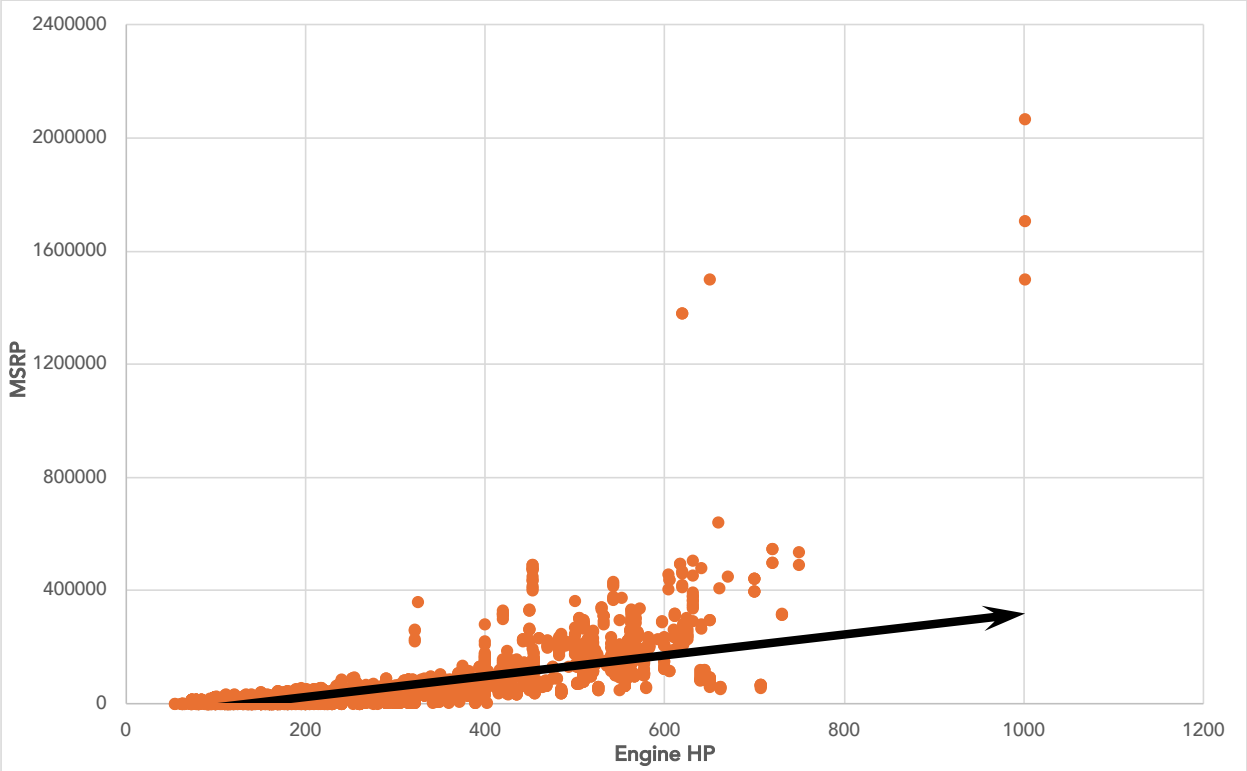
INSIGHT:

- N/a has high and diesel with less models.
- Hybrid has high and exotic has less popularity.

Insight Required: What is the relationship between a car's engine power and its price?

- **Task 2:** Create a scatter chart that plots engine power on the x-axis and price on the y-axis. Add a trendline to the chart to visualize the relationship between these variables.

ANS.



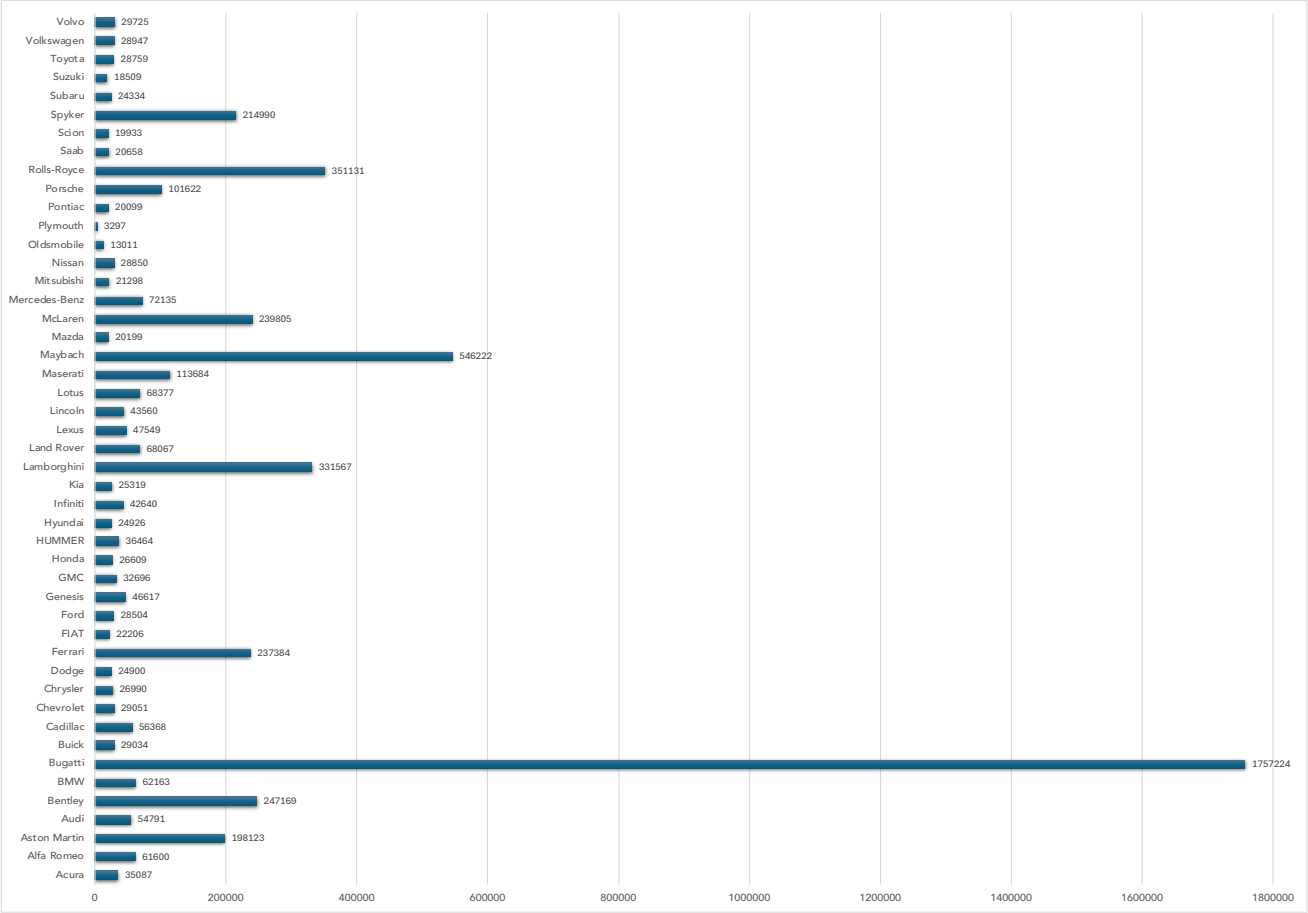
INSIGHT: As the horsepower of car's engine increases, slightly we see increase in car's cost.

Insight Required: How does the average price of a car vary across different manufacturers?

- **Task 4.A:** Create a pivot table that shows the average price of cars for each manufacturer.
- **Task 4.B:** Create a bar chart or a horizontal stacked bar chart that visualizes the relationship between manufacturer and average price.

ANS.

Brand	Average of MSRP
Acura	35087
Alfa Romeo	61600
Aston Martin	198123
Audi	54791
Bentley	247169
BMW	62163
Bugatti	1757224
Buick	29034
Cadillac	56368
Chevrolet	29051
Chrysler	26990
Dodge	24900
Ferrari	237384
FIAT	22206
Ford	28504
Genesis	46617
GMC	32696
Honda	26609
HUMMER	36464
Hyundai	24926
Infiniti	42640
Kia	25319
Lamborghini	331567
Land Rover	68067
Lexus	47549
Lincoln	43560
Lotus	68377
Maserati	113684
Maybach	546222
Mazda	20199
McLaren	239805
Mercedes-Benz	72135
Mitsubishi	21298
Nissan	28850
Oldsmobile	13011
Plymouth	3297
Pontiac	20099
Porsche	101622
Rolls-Royce	351131
Saab	20658
Scion	19933
Spyker	214990
Subaru	24334
Suzuki	18509
Toyota	28759
Volkswagen	28947
Volvo	29725

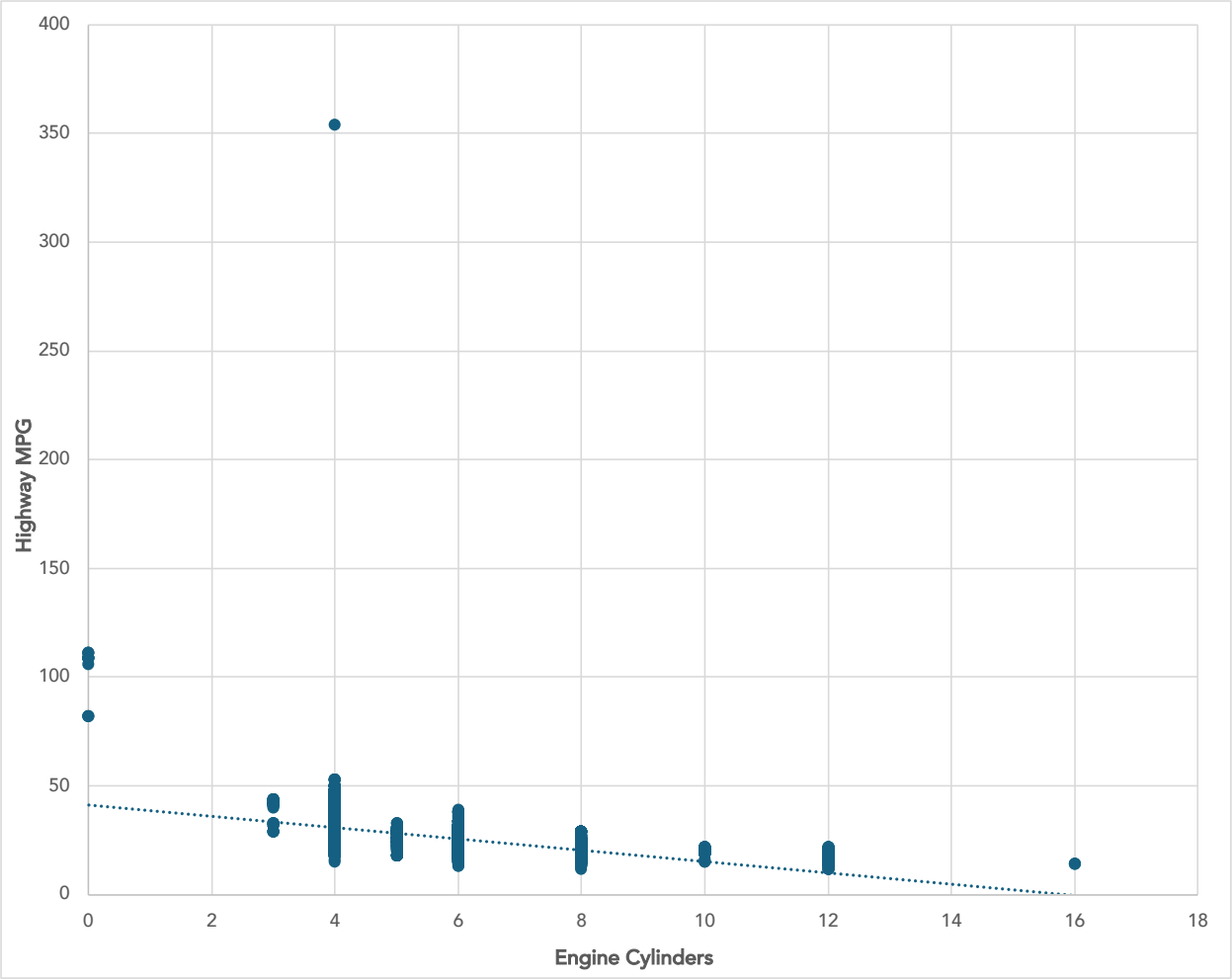


INSIGHT: Bugatti, Maybach, Rolls-Royce and Lamborghini offer expensive cars whereas Plymouth, Oldsmobile, Suzuki offer cheaper and affordable cars.

Insight Required: What is the relationship between fuel efficiency and the number of cylinders in a car's engine?

- Task 5.A: Create a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis. Then create a trendline on the scatter plot to visually estimate the slope of the relationship and assess its significance.
- Task 5.B: Calculate the correlation coefficient between the number of cylinders and highway MPG to quantify the strength and direction of the relationship.

ANS.



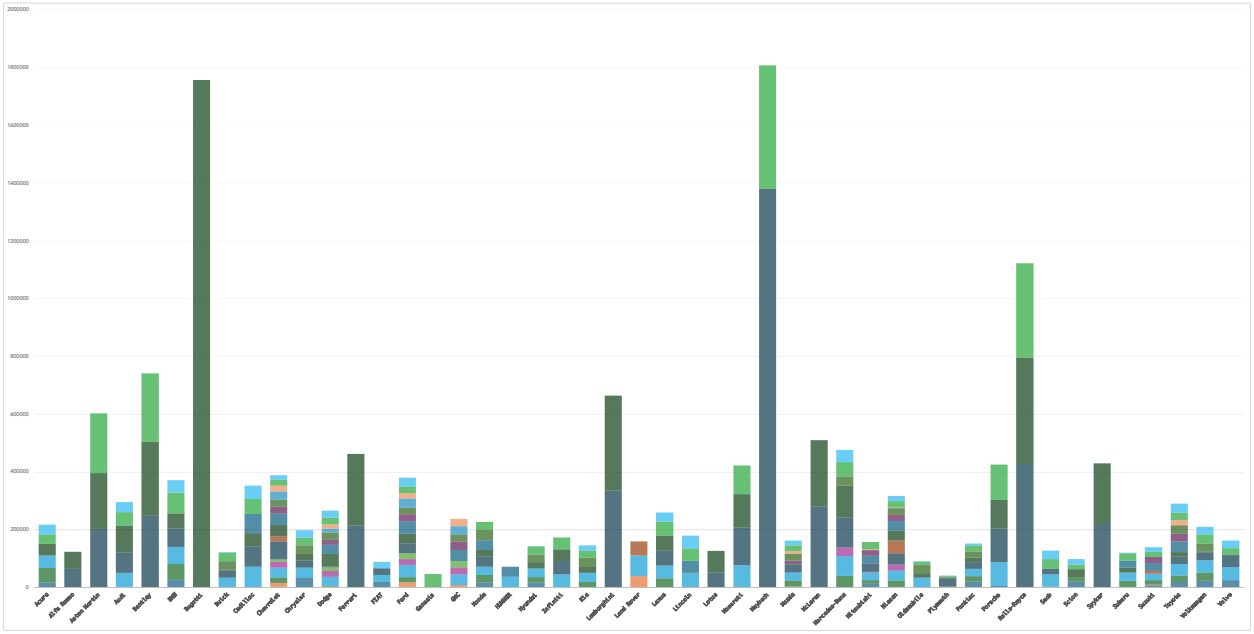
Correlation coefficient: **-0.615634959**

INSIGHT: Correlation coefficient tells that as cylinders (weight of the car) increases, fuel efficiency decreases. It is negatively correlated.

Task 1: How does the distribution of car prices vary by brand and body style?

ANS.

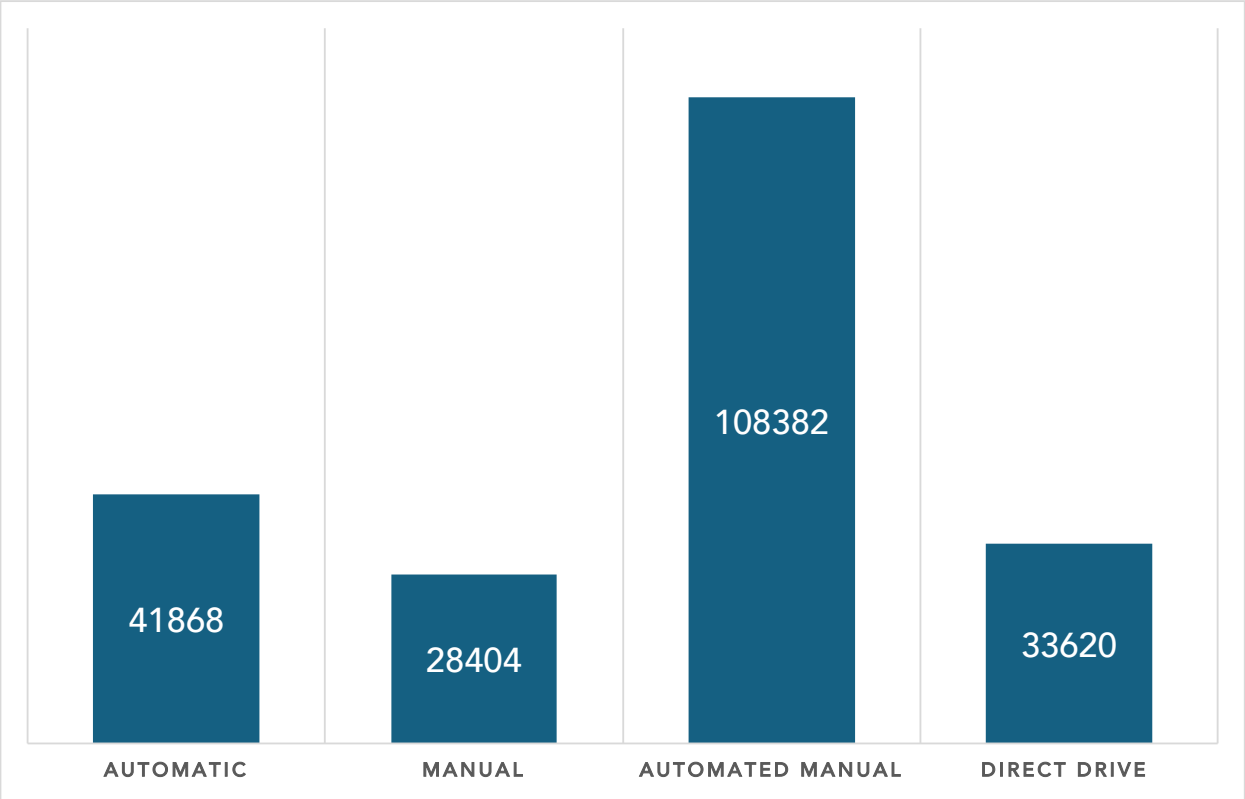
Average of MSRP Brand	Vehicle Style 2Dr Hatchback	2Dr Suv	4Dr Hatchback	4Dr Suv	Cargo Minivan	Cargo Van	Convertible	Convertible Suv	Coupe	Crew Cab Pickup	Extended Cab Pickup	Passenger Minivan	Passenger Van	Regular Cab Pickup	Sedan	Wagon
Acura	17176		51063	42960					39687						33614	33560
Alfa Romeo							64900		59400							
Aston Martin							203379		192893						206962	
Audi	2000			48635			70030		93587						46447	35223
Bentley							250536		254270						236836	
BMW	26699		55155	58536			63814		52445						71832	43267
Bugatti									1757224							
Buick				33996			25618		2059			30006			29569	2053
Cadillac				72551			70401		45440	66572					51179	47364
Chevrolet	2000	14716	18329	33717	20007	8299	62835	17717	38939	39256	24170	24934	28556	19825	19883	15825
Chrysler	32935			35792			25124		22502			29751			26104	26372
Dodge	2000	2000	2000	31176	20173	12537	2000		45058	31406	16302	25338	14142	15149	22519	24783
Ferrari							214719		248224							
FIAT	19136			24620			23426									22121
Ford	2000	16134	18468	42028	19700	20606	34762		34101	41991	23808	22587	32836	17798	23259	30066
Genesis															46617	
GMC		8488		37480	23792	21909				39062	27896	25105	28556	25183		
Honda	17217		26372	28576			36019		21763	34101		36879			26027	
HUMMER				37749						34629						
Hyundai	18364		17629	30218					22126			26615			27667	
Infiniti				45686			46669		40292						41076	
Kia			19379	31533					20376			32977			23812	18217
Lamborghini							336402		328292							
Land Rover		39700		71284				48577								
Lexus			31567	45042			52452		50824						48865	31105
Lincoln				50332					2168	41205					41665	44951
Lotus							51658		75065							
Maserati				77500			130165		116017						99022	
Maybach							1381375								426914	
Mazda	2000	2000	20809	27141			28081		2000		11601	23323		9155	19739	16675
McLaren							280225		229700							
Mercedes-Benz			40933	68400	28950		104618		109714			32500			48834	43069
Mitsubishi	12764		12879	26116	2000		29985			26690	19194	2000		2000	24058	
Nissan	2098		22241	34294	21437		39071	43692	35393	32736	20528	22962		2213	22604	17500
Oldsmobile				34021			2000		10961			32804			9238	2000
Plymouth	2000		2000				28544		2000			2113			2769	2000
Pontiac	21255		18108	25097			23196		16188			20815			20992	6952
Porsche	5765			82509			115502		99136						123341	
Rolls-Royce							428273		367446						326951	
Saab	2000		2034	41685			19802								33356	28855
Scion	20351		15693						27518						16250	18445
Spyker							219990		209990							
Subaru	2000		21189	28538					16785	24398					26187	2000
Suzuki	7416	2000	16697	21674				9350		27648	21638				17990	15539
Toyota	18950		22187	40632			25778		15615	36846	26251	30039		17593	24800	31742
Volkswagen	24135		28198	41699			27674		2000			29240			30796	26386
Volvo	26258			45387			40533		2000						22290	26271



Task 3: How do the different feature such as transmission type affect the MSRP, and how does this vary by body style?

ANS.

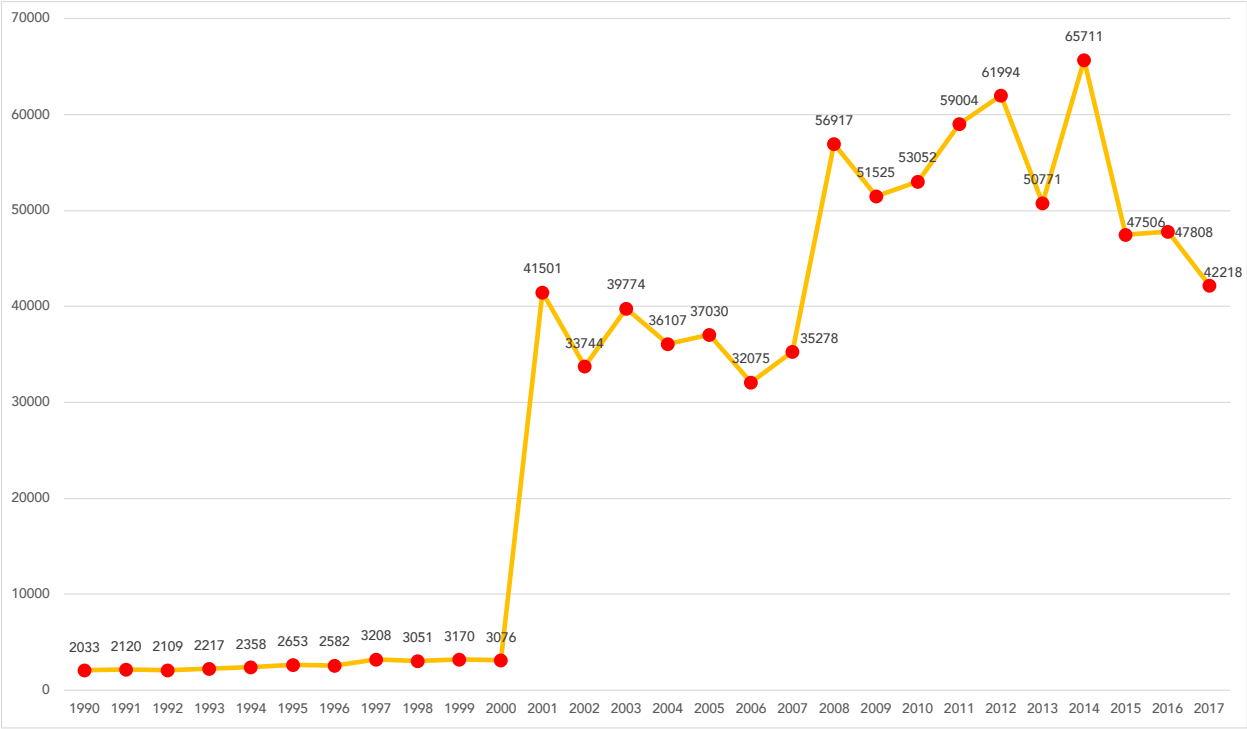
Transmission Type	Average of MSRP
Automatic	41868
Manual	28404
Automated Manual	108382
Direct Drive	33620
Grand Total	**42016**



Task 2: How does the fuel efficiency of cars vary across different body styles and model years?

ANS.

Year	Average of MSRP
1990	2033
1991	2120
1992	2109
1993	2217
1994	2358
1995	2653
1996	2582
1997	3208
1998	3051
1999	3170
2000	3076
2001	41501
2002	33744
2003	39774
2004	36107
2005	37030
2006	32075
2007	35278
2008	56917
2009	51525
2010	53052
2011	59004
2012	61994
2013	50771
2014	65711
2015	47506
2016	47808
2017	42218



INSIGHT: Before 2000, MSRP seems to have steady growth but after, it suddenly surges high and shows high volatility and reaches its peak in 2014. Later, it continues to drop.